

Toward an Ever-present Extended Reality: Distinguishing Between Real and Virtual

Toward an Ever-present Extended Reality

Anton Nijholt

Human Media Interaction, University of Twente, The Netherlands
a.nijholt@utwente.nl

ABSTRACT

In this short paper we provide some observations on the (im)possibility of distinguishing between real and virtual in augmented reality worlds. Distinguishing may be possible because of imperfect technology, impossibilities, and manipulating perceptual capabilities. The question must be asked, if this distinction cannot always be made, does this place responsibility on the owner or administrator of the Extended Reality (XR) environments? And who might be the owner or administrator in the case of ever-present extended reality?

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality.

KEYWORDS

Augmented Reality, Augmented Reality Glasses, Optical See-Through, Extended Reality, Multi-sensorial Virtual Content, Perception, Perceptual Manipulation, Virtual, Real

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1 INTRODUCTION

In this short paper, we present some observations on the (im)possibility of distinguishing between real and virtual in Extended Reality (XR) worlds. This paper builds on two earlier papers [1, 2]. In XR, virtual objects are superimposed on the real world. The virtual objects can be multisensorial. That is, they can address one or more of our senses, from vision to proprioception. As a starting point, we refer to Azuma's definition [3] of Augmented Reality (AR). AR combines real and virtual objects in a real environment (1), registers (aligns) real and virtual objects (2), and runs interactively, in three dimensions, and in real time (3).



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Azuma [4] predicted that AR technology, in particular with optical see-through glasses, will become the dominant platform and interface for accessing digital information, thus supplanting the smartphone. Hence, smart glasses are seen as "... the best chance of achieving the long-term vision of ubiquitous consumer AR displays." Smart glasses can be combined with wearables that allow us to sense information that is not visual, such as information that provides us with auditory, tactile, olfactory, or gustatory experiences or combinations of these experiences.

Looking at current technology, we can see that Azuma's definition should be extended. Not all virtual content is necessarily interactive. There can be static virtual content that does not change position in the augmented world and that cannot be manipulated. Moreover, there can be autonomous virtual content that moves around in the augmented world without user interference. This is one of the reasons that we prefer to talk about mixed reality (MR) or extended reality (XR). Throughout this paper, we assume that virtual objects are superimposed on reality.

2 DISTINGUISHING BETWEEN REAL AND VIRTUAL

Can we distinguish between virtual and real objects? Why should we call a taste or a smell virtual? Our taste or smell receptors do not register the difference. Neither do the receptors in our eyes. We distinguish between real and virtual—if we can distinguish at all—because the virtual is at odds with our common sense and our knowledge of the world. In this note that accompanies our keynote talk, we provide observations on being able to distinguish between what is virtual and what is real in XR. Although XR is often considered in terms of the sense of sight, content that is artificially added to an environment can appeal to any of our senses. Thus, further observations are already needed. Why should we call added content artificial? 'Artificial' content is already added to our food and drinks to change taste or for health concerns. When we enter a store we have to accept its music, and a brand fragrance may be diffused in some stores. Is that different from having artificial content displayed to our sense of sight? This paper cannot provide any final answer. Rather, we look at examples that may help to answer such a question.

One way to make this distinction is if we are aware that technology is needed to address one or more of our senses. Do we need to put a pill in our mouth to give us a particular taste experience? Is there a device that injects a smell into our nose? Is there a projection of visual content on a wall? Are we asked to put on special glasses to see virtual objects added to our physical environment? Our prescribed glasses also change our environment, as do hearing

aids and heart valves. We have become accustomed to devices that give our environments a different appearance than that with which we are objectively familiar. In the case of sight and hearing aids, we can say that it involves improving and enhancing signals from the environment rather than adding new information. However, what about a real-time translation while viewing a streaming video? As mentioned, we do not yet know how to answer such questions. However, hopefully, our questions and examples will help to obtain better answers to our research.

Here, we will provide three answers to our research question: How can we decide between real and virtual content in XR? These three answers are inferred from imperfect technology, recognizing impossible situations, and having access to technology that modifies our perceptual qualities.

2.1 Imperfect Technology

Imperfect technology allows us to see that generated virtual content differs from real-life content. We should mention that not everyone will be able to make this distinction. Can we recognize imperfect technology? In the case of AR, the augmentation can be too perfect. Rather than seeing a three-dimensional cube that fits naturally in the environment, we see a perfect mathematical object with colors and edges that are computer-generated but that do not meet our expectations of how such an object fits into a natural environment. There is a contradiction between the appearance of the object and the environment in which it appears. Imperfect technology shows an object that is not sufficiently aware of the context where it appears, of light rainfall, or of signs of age and weathering. Due to insufficient detailed modelling of the environment occlusion problems will appear. Especially in multisensorial AR we can expect to observe errors in the synchrony between modalities. Can we ever expect a virtual object in outdoor-AR to get wet to the point where we can not only see the raindrops on the virtual object, but also, using haptics, feel the wetness on the virtual object? If not, then we can conclude that the object is virtual and not real. How realistic can we model virtual rain? Many such examples can be found in [5]. See also [6], where we discuss such matters for virtual humans that may appear in our AR worlds. Moreover, in the case of AR, the device that allows us to experience augmentation is not perfect. For example, in the case of AR Head-Mounted Devices, the augmentation can be limited to part of our field of view, and we can see where the virtual content does not fit reality.

2.2 Impossibilities

Recognizing impossible situations in XR requires common sense knowledge of what is and is not possible. However, that is not enough. When walking around in real-life Manhattan and meeting a very realistic Marilyn Monroe struggling with her dress, it is good to know that she already died in 1982. What about meeting a dodo or a dinosaur? Perhaps we should mention ‘physical’ laws. Dead persons and extinct animals are no longer part of our reality. Meeting Marilyn cannot be real, she is dead. Perhaps we should mention that meeting her contradicts physical laws. However, does that mean that we should put Marilyn in the same category as, for example, a flying horse? This is not very satisfactory. However,

meeting a human that is not aging or an extinct animal does not comply with physical rules.

A virtual animal that represents an animal that appears in legends or fantasy (e.g., Disney figures) may or may not comply with physical laws, depending on the description in the legend. Perhaps there is a legend or fiction in which a unicorn can talk or fly. What information about the unicorn makes us say that they therefore do not comply with physical laws? Disney’s fantasy figure Dumbo can fly. We can say that this is physically impossible, but then we forget about his large ears that serve as wings. In a legend, horses can fly because their bodies have wings. What about Greek mythology? These observations make it useless to distinguish between fantasy figures that comply or do not comply with physical laws. In addition, what about people who can walk on water and turn it into wine? Perhaps we should include them in the category of flying horses or fantasy figures?

We can look at a very simple implementation of AR and add our observations to that as well. Suppose we walk around in an urban environment and our AR glasses inform us about the buildings we see. That is, text messages are displayed on our glasses. They float in the air, and they change position and orientation when we do. When we take off our AR glasses, although, why would we do that, we do not see the text messages anymore. They do not appear in real life. They are floating billboards. On the other hand, we can also read that information in our physical tourist booklet or on our smartphone. Should we judge reading through our AR glasses as less realistic than reading in those other media? Should we put them in a category of flying horses, fantasy figures, or Marilyn Monroe?

In AR, multisensorial virtual content can exist. That is, in the virtual layer that is superimposed on reality, we can have virtual content aligned with reality that addresses all of our senses: the five classical senses (sight, hearing, taste, smell, and touch), proprioception, balance, and others (e.g., pain and temperature). For this discussion, we limit ourselves to the classical senses.

Suppose a computer-generated cube is visualized in the virtual layer. In reality, a physical cube can be seen and touched, we hear a sound when tapping it, and taste it by licking it, it has a smell, it has a weight, and it has been affected by temperature and humidity. If the augmentation only addresses the sight sense, then in the nonaugmented real world, an object comparable to this virtual cube does not exist. From the point of view of the real world, it has become an impossible object. If we confine ourselves to the senses for which a virtual object has been modeled, then nevertheless we can speak of ‘realistic’ behavior, for example, complying with physical laws. If the cube floats through the air and disappears in a solid real-world object, then it does not comply with physical laws. More senses can be involved. Realistic behavior and compliance with specific laws can be decided relative to one or more senses, leaving open whether or not the virtual object is properly modeled for the other senses or not at all. An invisible virtual object in the virtual layer may allow realistic touch exploration. Again, from the real-world point of view, this is an impossible object.

We return to common sense. Animals and humans display behavior that can be explained by common sense. If it is raining in the real world, a virtual dog that is in our field of view should become wet. If not, we observe an impossibility. A cat will seek a dry place.

If not, it must be virtual. Regardless of which senses we are allowed to use to experience a virtual cat, if it does not behave like a cat, it is not a real cat. Dogs are affectionate, and cats are more autonomous, and we can also label human behavior as possible or impossible. For example, all humans lie and cheat sometimes.

Suppose we can implement a virtual human that addresses all our senses as being in a real human would. For example, if we ask it to stand on a virtual scale, the scale should give a plausible weight. If we ask it to stand on a real scale in the augmented world, then the real scale, knowing about this virtual human through its sensors, should also display the same weight. Suppose again that this ‘human’ implemented in the virtual layer can display plausible autonomous, cognitive, and emotional behavior. However, there is one exception: it cannot lie and deceive. This makes it a different animal than a human. You could say that in these species a human trait is missing, or, viewed another way, a nonhuman trait (complete honesty) has been added. Where does it fit? Saying that it does not comply with physical laws leads us into a morass of philosophical discussions without reaching an answer. Then, perhaps we should call it a fantasy figure introduced by the author of the AR world. If so, any virtual animal species that does not have all of its real-life properties simulated should be called a fantasy figure. However, having all of them in the same class as unicorns or Mickey Mouse is again rather unsatisfactory. Humanoids can play useful roles in the AR (and XR) world. Nevertheless, this view is a possibility. Recall Mary Shelley’s book on the Frankenstein humanoid or the legend of Rabbi Lev’s Golem in Prague. Perhaps we can find fiction in which someone is introduced who is inhumanly honest. In fact, we can. In “Machines Like Me” [7], Ian McEwan introduces a robot that has these characteristics. It belongs to a species that does not exist, although it complies with physical laws. In the book, it is killed by its ‘owner’. Distinguishing our completely honest virtual human from a real human looking at appearance and behavior and trusting our senses is not enough.

2.3 Manipulating and Modifying our Perceptual Qualities

We think we know about our perceptual limitations. Technology allows us to use prescription glasses or hearing aids. The internet, our computers, smartphones, and social media allow us to contact, see and hear our loved ones wherever they are. We have become accustomed to it and do not see this as an impossibility, while perhaps some decades ago we would have considered this impossible. We may wonder if XR will provide a similar experience if it is ever present. Returning to the augmented world and our senses, if there is virtual content superimposed on the real world, can we distinguish between virtual and real?

Whether or not they are integrated with smart glasses, interconnected wearable devices make it possible to offer XR users perceptual possibilities that cannot be obtained without these technological devices, but users can become accustomed to them, as they have been accustomed to hearing aids and smartphones. Sensor information, wherever these sensors are located, can inform and guide a user wherever they are. The information can be displayed on a wearable XR device in a multisensorial way. Distant cameras and microphones, perhaps built into drones, can provide

information about places to enter. This can include information from, for example, infrared cameras or microphones that detect sound beyond the human ear, which is then visualized on a user’s XR device. In this way, we add “virtual skills” to our perceptual skills. Hence, we can mention magnification for low-vision users, seeing images that are obtained from the rear, having a 360° view, seeing images from remote and drone cameras on smart glasses, having an X-ray (see-through) vision, having nonvisual information visualized (for example, visualizing sound frequencies beyond the human ear), obtaining an infrared thermography camera view, integrating or shifting views (for example, from egocentric to exocentric), physically manipulating out-of-view objects, using smell sensors that enhance or visualize smell, or altering the speed of reality [8].

We do not think that adding these qualities and skills to deal with them will be seen as having access to virtual information. The reality is not augmented, and the way the reality is experienced is augmented by devices that offer users perception and perceptual skills that are not available without these devices. However, we will become accustomed to it and incorporate it into our daily activities.

3 WHO OWNS/MANIPULATES OUR (EXTENDED) REALITY?

In the observations we have provided above, the reader should be aware that, at least presently, video see-through AR rather than optical see-through AR is needed for the research examples that are mentioned.

In Diminished Reality (DR) [9], the reality that we see through our AR glasses is not augmented with virtual objects; instead, real objects are removed from reality. Different techniques can be used to realize this, but what they have in common is that a plausible and aligned background for the object is used to replace the object. This is not essentially different from augmenting an environment with virtual objects. Additionally, in that case, we have to take care that a virtual object is aligned with the environment and occludes (or is occluded by) real objects. Instead of a diminished reality where an object is removed from reality, we can also have a situation in which a real object is replaced by a virtual object. Why not offer a view of reality where objects we do not like are replaced by objects we like? This was already investigated by Steve Mann [10], who experimented with computer vision and an AR in which real-life billboard content was replaced with content that was more to a user’s preferences. The user creates their own reality, or the author/owner of the AR environment creates it for the user. The user is not necessarily aware of this. This is not different from a reality created for the user in social media based on their preferences but driven by commercial interests. However, now the created and experienced reality can be multisensorial.

It is useful to mention a patent filed by Apple researchers [11], who described plans to remove virtual humans that represent remote people from an AR view if they breach social interaction criteria. In this patent, they can be made partially invisible, fully invisible, or anywhere in between along a spectrum of attenuation. We or someone else can decide to erase such a person and their activities from our view. In this way, our augmented world is manipulated instead of our real-world view. Computer vision allows

us to detect and follow human beings that can be displayed in a blurred and audibly muffled way - as in the Sci-Fi “Black Mirror” series - on our AR device [12]. In a diminished reality view, technology should make it possible to replace silhouettes with their backgrounds while moving around. Cameras can also provide an AR user with images that contain only the information in which they or the AR environment owner are interested [13].

4 CONCLUSIONS

We have already alluded to it, but of course, all this also requires an answer to the question of who is responsible for the AR content provided to us. Is it television companies that are going to provide a more than tenuous offering? Or is it energy companies that are going to provide water, gas electricity, and entertainment? Who is the company or author of the AR environment where you perform your daily activities? How visible are you in your use of public transport, your presence in public areas, or your shopping activities? Moreover, what is the multisensorial information that can address your senses and create your augmented multisensorial world, whether you are aware of it or not?

It is useful to cite AR pioneer Louis Rosenberg [14]:

“After all, the shared experience we call “civilized society” is quickly eroding, largely because we each live in our own data bubble, everyone being fed custom news and information tailored to their personal beliefs, reinforcing their biases and entrenching their opinions. ... But today we can at least enter a public space and have some level of shared experience in a common reality. With AR, that too may be lost.”

Moreover,

“Third parties can not only inject their virtual content into AR but also filter out real-world content that does not meet a reality desired by stakeholders (a government, an international, national, or local company, or a consumer).”

Currently, much attention is being paid to AI-generated text, images and videos. Mostly we are confronted with them on social media. Judging from the citations above, it makes sense to extend this attention to our future XR experiences.

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Section 2.3 is a (very) short summary of [1]. A preliminary version of this paper is [2]. Observations on impossible objects in the context of human-food interaction appear in [15].

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